DSA0405-Fundamental of Data Science

CO3-AT1 - Case Study

UNIT -3

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Case Study 6 — Detecting an Unusual Customer:

A shopping website finds that the average amount spent by customers is ₹2,000, with a standard deviation of ₹400. A particular customer spends ₹3,200.

SOLUTION:

Given Data:

Mean spending, μ = ₹2,000

Standard deviation, σ = ₹400

Spending of the particular customer, x = ₹3,200

<u>Question 1: Calculate the customer's z-score.</u>

$z = (x - \mu) / \sigma$

Where:

z = the z-score (number of standard deviations a value is from the mean)

x = the observed data value = 3200

μ = the population mean = 2000

σ = the population standard deviation = 400

Step 1: Find the numerator $(x - \mu)$.

$x - \mu = 3200 - 2000 = 1200$

Step 2: Divide the numerator by σ.

$z = 1200 / 400$

Step 3: Simplify the division.

$z = 3$

Final Answer: z = 3

Question 2: Is the customer's spending unusually high compared with the average?

Rule: If |z| > 2 (or 3), the value is considered statistically unusual.

Step 1: Take the absolute value of the z-score calculated above.

$|z| = |3| = 3$

Step 2: Compare $|z|$ with the threshold value 2 (commonly used cut-off for 'unusual').

$3 > 2 \rightarrow$ condition satisfied

Final Answer: Yes, the spending of ₹3,200 is unusually high, since z = 3 places it 3 standard deviations above the mean.

Question 3: What would a negative z-score mean in this context?

$z = (x - \mu) / \sigma$

Step 1: Set the z formula less than 0 and solve the inequality for x.

$(x - \mu) / \sigma < 0$

Step 2: Since σ (400) is always positive, the sign of z depends only on $(x - \mu)$.

$(x - \mu) < 0 \;\Rightarrow\; x < \mu$

Step 3: Substitute $\mu = 2000$.

$x < 2000$

Final Answer: A negative z-score would mean the customer spent less than the average of ₹2,000.

Question 4: If another customer has a z-score of −1.5, what does it tell you?

$x = \mu + z \cdot \sigma$

Where:

- x = spending amount to be found
- μ = mean = 2000
- z = given z-score = −1.5
- σ = standard deviation = 400

Step 1: Multiply z by σ.

$z \times \sigma = -1.5 \times 400 = -600$

Step 2: Add this result to μ.

$x = 2000 + (-600)$

Step 3: Simplify.

$x = 1400$

Final Answer: The customer spent ₹1,400, which is 1.5 standard deviations below the mean — lower than typical spending, but not extreme enough to be called an outlier ($|z| < 2$).

Case Study 7 — Delivery Time Analysis

A delivery company records delivery times (in minutes): 25, 27, 28, 29, 30, 30, 31, 32, 34, 60

SOLUTION:

Data (delivery times in minutes), $n = 10$:

25, 27, 28, 29, 30, 30, 31, 32, 34, 60

Question 1: Calculate the mean and median.

Mean = Σxi / n

Where:

Σxi = sum of all data values

n = number of data values = 10

Step 1: Add the values in pairs to find Σxi.

$25 + 27 = 52$

$52 + 28 = 80$

$80 + 29 = 109$

$109 + 30 = 139$

$139 + 30 = 169$

$169 + 31 = 200$

$200 + 32 = 232$

$232 + 34 = 266$

$266 + 60 = 326$

Step 2: Divide the total by n.

Mean = 326 / 10 = 32.6

Final Answer: Mean = 32.6 minutes

Median (n even) = [(n/2)-th value + (n/2 + 1)-th value] / 2

Step 1: n = 10, so n/2 = 5th value and n/2+1 = 6th value.

5th value = 30, 6th value = 30

Step 2: Average the two middle values.

Median = (30 + 30) / 2 = 60 / 2 = 30

Final Answer: Median = 30 minutes

Question 2: Find Q1 and Q3.

Q1 = median of the lower half of data; Q3 = median of the upper half of data

Step 1: Split the ordered data into two halves of 5 values each (excluding overall median position).

Lower half: 25, 27, 28, 29, 30

Upper half: 30, 31, 32, 34, 60

Step 2: Find the median of the lower half (middle of 5 values) → Q1.

Lower half ordered: 25, 27, 28, 29, 30 → middle value = 28

Final Answer: Q1 = 28 minutes

Step 3: Find the median of the upper half (middle of 5 values) → Q3.

Upper half ordered: 30, 31, 32, 34, 60 → middle value = 32

Final Answer: Q3 = 32 minutes

Question 3: Calculate the IQR.

IQR = Q3 − Q1

Step 1: Substitute Q3 = 32 and Q1 = 28.

IQR = 32 − 28

Step 2: Simplify.

IQR = 4

Final Answer: IQR = 4 minutes

Question 4: Check for outliers using the 1.5 × IQR rule.

$$\textit{Lower Fence} = Q1 - 1.5 \times IQR \qquad \textit{Upper Fence} = Q3 + 1.5 \times IQR$$

Step 1: Compute 1.5 × IQR.

$1.5 \times \text{IQR} = 1.5 \times 4 = 6$

Step 2: Compute the lower fence.

$\text{Lower Fence} = 28 - 6 = 22$

Step 3: Compute the upper fence.

$\text{Upper Fence} = 32 + 6 = 38$

Step 4: Compare every data value against the fences [22, 38].

All values 25–34 lie inside [22, 38]. The value 60 lies outside ($60 > 38$).

Final Answer: 60 is an outlier; no other values are outliers.

Question 5: Explain whether the mean or median gives a better description of the typical delivery time.

Step 1: Compare mean and median computed above.

Mean = 32.6, Median = 30

Step 2: Note the effect of the outlier (60) on each measure.

The outlier raises the mean above most of the data ($32.6 > 8$ of the 10 values), while the median (30) stays close to the bulk of the data.

Final Answer: The median (30 minutes) better represents the typical delivery time, since it is not distorted by the outlier.

Case Study 8 — Two Cities' Temperatures

The average daily temperatures (°C) for 7 days are:

City A: 28, 29, 30, 30, 31, 30, 32

City B: 20, 25, 30, 35, 30, 25, 45

SOLUTION:

Given Data

City A (°C), $n = 7$: 28, 29, 30, 30, 31, 30, 32

City B (°C), $n = 7$: 20, 25, 30, 35, 30, 25, 45

Question 1: Calculate the mean temperature for each city.

Mean = $\Sigma xi / n$

Step 1 (City A): Add all 7 values.

$28 + 29 + 30 + 30 + 31 + 30 + 32 = 210$

Step 2 (City A): Divide by $n = 7$.

Mean(A) = $210 / 7 = 30$

Final Answer: Mean of City A = 30°C

Step 1 (City B): Add all 7 values.

$20 + 25 + 30 + 35 + 30 + 25 + 45 = 210$

Step 2 (City B): Divide by $n = 7$.

Mean(B) = $210 / 7 = 30$

Final Answer: Mean of City B = 30°C

Question 2: Calculate the variance for each city.

$\sigma^2 = \Sigma(xi - \mu)^2 / n$

Where:

xi = each individual data value

μ = the mean of the data set

$(x_i - \mu)$ = deviation of each value from the mean

n = number of data values

Step 1 (City A): Find the deviation and squared deviation of each value from $\mu = 30$.

x_i	$x_i - \mu$	$(x_i - \mu)^2$
28	-2	4
29	-1	1
30	0	0
30	0	0
31	1	1
30	0	0
32	2	4

Step 2 (City A): Sum the squared deviations.

$\Sigma(x_i - \mu)^2 = 4+1+0+0+1+0+4 = 10$

Step 3 (City A): Divide by $n = 7$.

$\sigma^2(A) = 10 / 7 = 1.43$

Final Answer: Variance of City A = 1.43

Step 1 (City B): Find the deviation and squared deviation of each value from $\mu = 30$.

x_i	$x_i - \mu$	$(x_i - \mu)^2$
20	-10	100
25	-5	25
30	0	0
35	5	25
30	0	0
25	-5	25
45	15	225

Step 2 (City B): Sum the squared deviations.

$\Sigma(x_i - \mu)^2 = 100+25+0+25+0+25+225 = 400$

Step 3 (City B): Divide by $n = 7$.

$\sigma^2(B) = 400 / 7 = 57.14$

Final Answer: Variance of City B = 57.14

Question 3: Which city has greater temperature variability?

Step 1: Compare the two variances calculated above.

$\sigma^2(A) = 1.43$ vs $\sigma^2(B) = 57.14$

Step 2: The larger variance indicates greater spread of data around the mean.

$57.14 > 1.43$

Final Answer: City B has far greater temperature variability.

Question 4: Why can two datasets have the same mean but different standard deviations?

Mean $= \Sigma x_i / n$ ***(location measure)*** $\sigma = \sqrt{[\Sigma(x_i - \mu)^2 / n]}$ ***(spread measure)***

Step 1: Note that both formulas use different information from the data.

Mean uses only the sum of values; standard deviation uses the squared distance of each value from the mean.

Step 2: Apply this to City A and City B: both have $\Sigma x_i = 210$ giving $\mu = 30$, but the individual values differ greatly in how far they lie from 30.

City A values stay within ± 2 of 30; City B values range from -10 to $+15$ relative to 30.

Final Answer: Two datasets can share the same mean (average level) while having different standard deviations (different spread), because the mean and standard deviation measure different properties of the data.

Question 5: Which city has more predictable temperatures?

Step 1: Predictability corresponds to lower variability, i.e. lower σ^2.

$\sigma^2(A) = 1.43$ is much smaller than $\sigma^2(B) = 57.14$

Final Answer: City A has more predictable temperatures.

Case Study 9 — Data Center CPU Usage

A data scientist records CPU usage percentages: 40, 42, 45, 45, 46, 48, 50, 52, 55, 90

SOLUTION:

Data (CPU usage %), n = 10:

40, 42, 45, 45, 46, 48, 50, 52, 55, 90

Question 1: Calculate mean, median, and mode.

Mean = Σxi / n

Step 1: Add the values in pairs.

40 + 42 = 82

82 + 45 = 127

127 + 45 = 172

172 + 46 = 218

218 + 48 = 266

266 + 50 = 316

316 + 52 = 368

368 + 55 = 423

423 + 90 = 513

Step 2: Divide by n = 10.

Mean = 513 / 10 = 51.3

Final Answer: Mean = 51.3

Median (n even) = [(n/2)-th value + (n/2 + 1)-th value] / 2

Step 1: n = 10, so use 5th and 6th values: 46 and 48.

Step 2: Average them.

Median = (46 + 48) / 2 = 94 / 2 = 47

Final Answer: Median = 47

Step 1: Mode = the value that occurs most frequently.

45 occurs twice; all other values occur once.

Final Answer: Mode = 45

Question 2: Calculate the range.

Range = Maximum value − Minimum value

Step 1: Identify maximum and minimum.

Maximum = 90, Minimum = 40

Step 2: Subtract.

Range = 90 − 40 = 50

Final Answer: Range = 50

Question 3: Find Q1, Q3 and IQR.

Q1 = median of lower half; Q3 = median of upper half; IQR = Q3 − Q1

Step 1: Split data into lower and upper halves of 5 values each.

Lower half: 40, 42, 45, 45, 46

Upper half: 48, 50, 52, 55, 90

Step 2: Find the middle value of each half.

Q1 = middle of lower half = 45

Q3 = middle of upper half = 52

Step 3: Compute IQR.

IQR = Q3 − Q1 = 52 − 45 = 7

Final Answer: Q1 = 45, Q3 = 52, IQR = 7

Question 4: Identify possible outliers.

Lower Fence = Q1 − 1.5 × IQR   Upper Fence = Q3 + 1.5 × IQR

Step 1: Compute 1.5 × IQR.

$1.5 \times IQR = 1.5 \times 7 = 10.5$

Step 2: Compute the lower fence.

Lower Fence = $45 - 10.5 = 34.5$

Step 3: Compute the upper fence.

Upper Fence = $52 + 10.5 = 62.5$

Step 4: Compare each data value with the fences [34.5, 62.5].

All values 40–55 lie inside the fences. 90 lies outside ($90 > 62.5$).

Final Answer: 90 is an outlier.

Question 5: Determine the direction of skewness.

If Mean > Median → right-skewed (positive skew); If Mean < Median → left-skewed (negative skew)

Step 1: Compare mean and median found above.

Mean = 51.3, Median = 47

Step 2: Since Mean > Median, apply the right-skew rule.

$51.3 > 47$

Final Answer: The data is right-skewed (positively skewed), caused by the high outlier value 90.

Case Study 10 — Comparing Two Datasets

Two data analysts receive the following datasets:

Dataset A: 10, 20, 30, 40, 50

Dataset B: 28, 29, 30, 31, 32

SOLUTION:

Given Data

Dataset A, n = 5: 10, 20, 30, 40, 50

Dataset B, n = 5: 28, 29, 30, 31, 32

<u>Question 1: Calculate the mean of both datasets.</u>

Mean = Σxi / n

Step 1 (Dataset A): Add all values.

10 + 20 + 30 + 40 + 50 = 150

Step 2 (Dataset A): Divide by n = 5.

Mean(A) = 150 / 5 = 30

Final Answer: Mean of Dataset A = 30

Step 1 (Dataset B): Add all values.

28 + 29 + 30 + 31 + 32 = 150

Step 2 (Dataset B): Divide by n = 5.

Mean(B) = 150 / 5 = 30

Final Answer: Mean of Dataset B = 30

<u>Question 2: Calculate the variance of both.</u>

$\sigma^2 = \Sigma(x_i - \mu)^2 / n$

Step 1 (Dataset A): Find deviation and squared deviation of each value from μ = 30.

xi	xi − μ	(xi − μ)²

10	-20	400
20	-10	100
30	0	0
40	10	100
50	20	400

Step 2 (Dataset A): Sum the squared deviations.

$\Sigma(xi - \mu)^2 = 400+100+0+100+400 = 1000$

Step 3 (Dataset A): Divide by n = 5.

$\sigma^2(A) = 1000 / 5 = 200$

Final Answer: Variance of Dataset A = 200

Step 1 (Dataset B): Find deviation and squared deviation of each value from $\mu = 30$.

xi	**xi − μ**	**(xi − μ)²**
28	-2	4
29	-1	1
30	0	0
31	1	1
32	2	4

Step 2 (Dataset B): Sum the squared deviations.

$\Sigma(xi - \mu)^2 = 4+1+0+1+4 = 10$

Step 3 (Dataset B): Divide by n = 5.

$\sigma^2(B) = 10 / 5 = 2$

Final Answer: Variance of Dataset B = 2

Question 3: Calculate the standard deviation of both.

$\sigma = \sqrt{\sigma^2}$

Step 1 (Dataset A): Take the square root of the variance.

$\sigma(A) = \sqrt{200} = 14.14$

Final Answer: SD of Dataset A = 14.14

Step 1 (Dataset B): Take the square root of the variance.

$\sigma(B) = \sqrt{2} = 1.41$

Final Answer: SD of Dataset B = 1.41

Question 4: Which dataset has greater spread?

Step 1: Compare the two standard deviations.

$\sigma(A) = 14.14 \quad \text{vs} \quad \sigma(B) = 1.41$

Step 2: The larger standard deviation indicates greater spread.

$14.14 > 1.41$

Final Answer: Dataset A has far greater spread.

Question 5: If both datasets represent model errors, which model appears more consistent?

Step 1: Consistency corresponds to lower standard deviation (less variation in errors).

$\sigma(B) = 1.41$ is much smaller than $\sigma(A) = 14.14$

Final Answer: The model represented by Dataset B is more consistent.

Question 6: Explain why mean alone is not enough to compare datasets.

Step 1: Compare the means of the two datasets.

$\text{Mean}(A) = 30 = \text{Mean}(B)$

Step 2: Compare the standard deviations of the two datasets.

$\sigma(A) = 14.14 \neq \sigma(B) = 1.41$

Step 3: Conclude what the mean fails to capture.

Although both datasets have identical means, their variability is completely different, so mean alone cannot distinguish a highly-spread dataset from a tightly-clustered one.

Final Answer: Mean alone is not enough; variance/standard deviation must also be examined to fully compare two datasets.